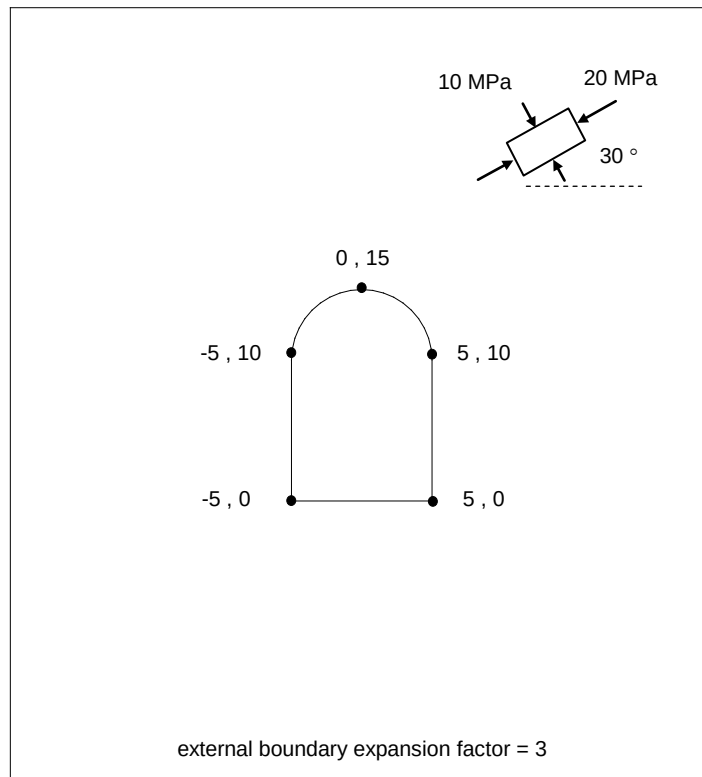


Quick Start Tutorial



This “quick start” tutorial will demonstrate some of the basic features of *Phase2* using the simple model shown above. You will see how quickly and easily a model can be created and analyzed with *Phase2*.

The finished product of this tutorial can be found in the **Tutorial 01 Quick Start.fez** data file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Model

If you have not already done so, run the *Phase2* Model program by double-clicking on the *Phase2* icon in your installation folder. Or from the Start menu, select Programs → Rocscience → Phase2 8.0 → Phase2.

If the *Phase2* application window is not already maximized, maximize it now, so that the full screen is available for viewing the model.

Note that when the *Phase2* Model program is started, a new blank document is already opened, allowing you to begin creating a model immediately.

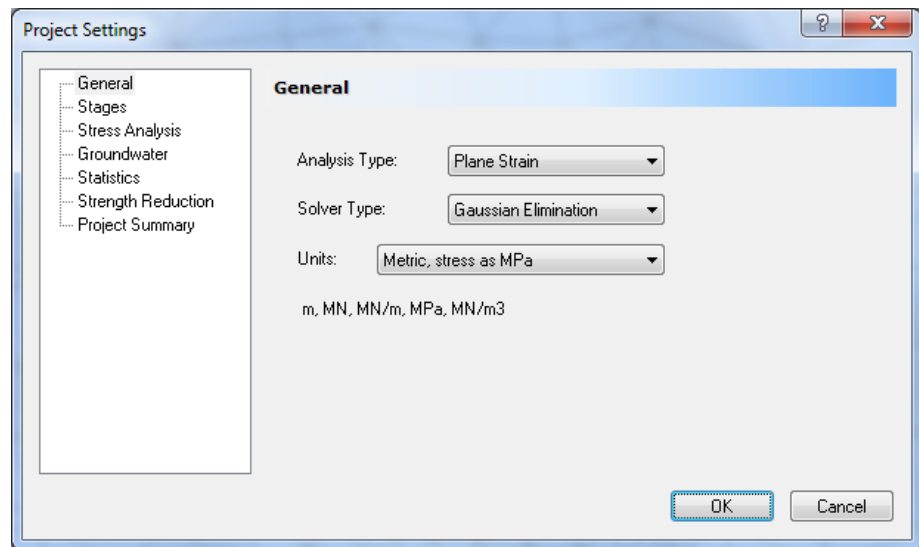
Project Settings

The Project Settings dialog is used to configure the main analysis parameters for your *Phase2* model.

Although we do not need to customize the Project Settings for this tutorial, let's take a look at the dialog.



Select: Analysis → Project Settings



Since we will be using Metric (MPa) units for this tutorial, you should make sure that the Units option is set to Metric, stress as MPa, under the **General** tab. This determines the units of length, force, stress and unit weight used in the analysis.

NOTE: *Phase2* remembers the most recently selected Units in Project Settings, and uses this as the setting for all new documents.

Select the **Stages** tab, and note that we will be analyzing a single stage model. (Multi-stage modeling is covered in the next tutorial).

Select the **Project Summary** tab and enter *Quick Start Tutorial* as the Project Title.

Do not change any other settings in the dialog. Select OK.

Entering Boundaries

First create the excavation as follows:



Select: Boundaries → Add Excavation

Enter the following coordinates in the prompt line at the bottom right of the screen. Note: press *Enter* at the end of each line, to enter each coordinate pair, or single letter text command (e.g. “a” for arc).

```
Enter vertex [t=table,i=circle,esc=cancel]: -5 10
Enter vertex [t=table,a=arc,i=circle,u=undo,esc=cancel]: -5 0
Enter vertex [...]: 5 0
Enter vertex [...]: 5 10
Enter vertex [...]: a
You will see the Arc Options dialog.
Select the 3 Points on Arc option, Number of Segments = 20.
Select OK.
Enter second arc point [u=undo,esc=cancel]: 0 15
Enter third arc point [u=undo,esc=cancel]: c
```

By entering “c” at the last prompt, the arc closes on the first point of the excavation. Note that arcs in *Phase2* are actually made up of a series of straight line segments. The Arc option and many other useful shortcuts are also available in the right-click menu.



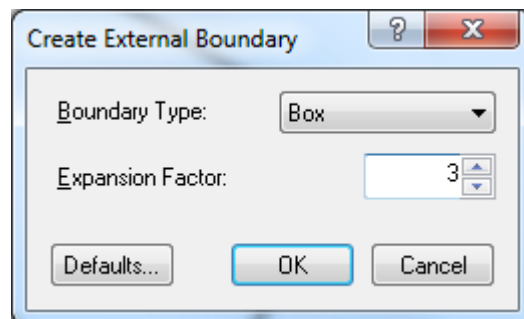
Select Zoom All (or press the F2 function key) to zoom the excavation to the center of the view.

Now we will create the external boundary. In *Phase2*, the external boundary may be automatically generated, or user-defined. We will use one of the ‘automatic’ options.



Select: Boundaries → Add External

You will see the Create External Boundary dialog. We will use the default settings of Boundary Type = Box and Expansion Factor = 3, so just select OK, and the external boundary will be automatically created.



The boundaries for this example have now been entered.

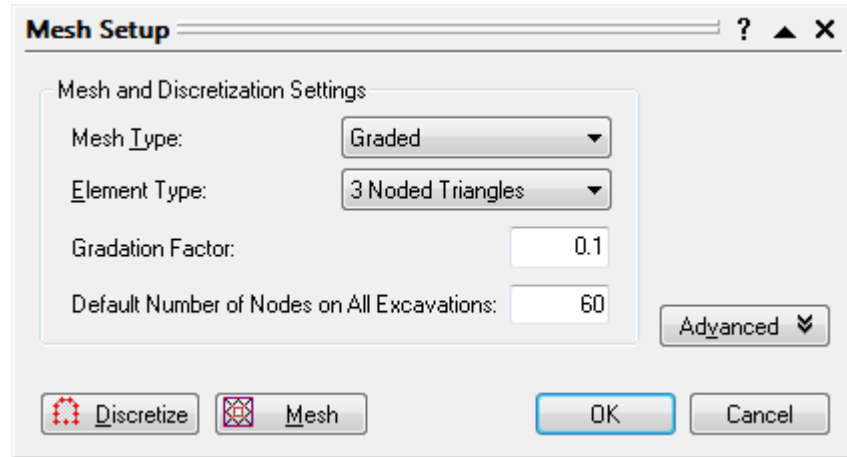
Meshing

The next step is to generate the finite element mesh. In *Phase2*, meshing is a simple two-step process. First you must Discretize the boundaries, and then the Mesh can be generated. You can also configure various Mesh Setup parameters before generating the mesh. We will do this first, although default parameters are in effect if you do not use the Mesh Setup option.



Select: Mesh → Mesh Setup

✓ Enter:
 Mesh Type = Graded
 Elem. Type = 3 Noded Tri.
 Gradation Factor = 0.1
 ✓ # Excavation Nodes = 60



Enter the # of Excavation Nodes = 60, and select OK.

Now discretize the boundaries.



Select: Mesh → Discretize

The discretization of the boundaries, indicated by red crosses, will form the framework for the finite element mesh. Notice the summary of discretization shown in the status bar, indicating the actual number of discretizations for each boundary type.

Discretizations: Excavation=59 External=49

Note that the number of excavation discretizations is 59, but we entered 60 in the Mesh Setup dialog. Don't worry, this is normal. Due to the nature of the discretization process, the actual number will not always be the same as the number you entered. If you are not happy with a given discretization, it can always be customized using a variety of custom discretization options (see the *Phase2* help topics for more information).

Now generate the finite element mesh, by selecting the Mesh option from the toolbar or the Mesh menu.



Select: Mesh → Mesh

The finite element mesh is generated, with no further intervention by the user. When finished, the status bar will indicate the number of elements and nodes in the mesh:

ELEMENTS (EL) = 634 NODES (ND) = 371

If you have followed the steps correctly so far, you should get the same number of nodes and elements as indicated above.

TIP: you can Discretize and Mesh in one step, by selecting the **Discretize and Mesh** option.

Boundary Conditions

For this tutorial, no boundary conditions need to be specified by the user. The default boundary condition will therefore be in effect, which is a fixed (i.e. zero displacement) condition for the external boundary.

Field Stress

Field Stress determines the initial in-situ stress conditions, prior to excavation.

In *Phase2* you can define either a Constant field stress or a Gravity field stress. For this tutorial we will use a Constant field stress.



Select: Loading → Field Stress

- ✓ Enter:
- Fld. Str. Type = Constant
- ✓ Sigma 1 = 20
- Sigma 3 = 10
- Sigma Z = 10
- ✓ Angle = 30

Enter *Sigma 1* = 20, *Angle* = 30, and select *OK*.



Notice that the small “stress block” in the upper right corner of the view indicates the relative magnitude and orientation of the field stress you entered. *Note the definition of the Constant Field Stress Angle in Phase2 – the Angle is the counter-clockwise angle between the Sigma 1 direction and the horizontal axis.*

Properties

We will now define the properties of the rock mass.



Select: Properties → Define Materials

With the first tab selected, enter the following properties:

✓ Enter:

- ✓ Name = rock mass
- Init.El.Ld.=Fld Stress Only
- Material Type = Isotropic
- Young's Modulus = 20000
- ✓ Poisson's Ratio = 0.2
- Failure Crit. = Mohr Coul.
- Material Type = Elastic
- Tens. Strength = 0
- Fric. Angle (peak) = 35
- ✓ Cohesion (peak) = 12

Enter Name = rock mass, Poisson's ratio = 0.2 and cohesion = 12 MPa, and select OK.

Since you entered properties with the first (Material 1) tab selected, you do not have to Assign these properties to the model. *Phase2* automatically assigns the Material 1 properties for you.

If you define properties with the Material 2, Material 3, Material 4 etc. tabs (e.g. for a multiple material model), then you will have to use the Assign option to assign these properties. We will deal with assigning properties in Tutorial 2.

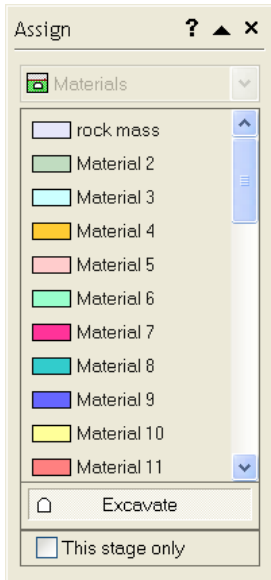
Excavating

We have one last thing to do to complete our simple model. Although we do not have to assign material properties, we do have to use the Assign Properties option, in order to excavate the material from within the excavation boundary. This is easily done with a few mouse clicks.



Select: Properties → Assign Properties

You will see the Assign Properties dialog, shown in the margin.



1. Use the mouse to select the Excavate button at the bottom of the Assign Properties dialog.
2. A small cross-hair icon (+) will appear at the end of the cursor. Place the cross-hair anywhere within the excavation boundary, and click the left mouse button.
3. The elements within the excavation boundary will disappear, indicating that the region within the boundary is now “excavated”.
4. That is all that is required. Select the **X** button at the upper right corner of the Assign dialog (or press Escape twice, once to exit the “excavate” mode, and once to close the dialog). The Assign dialog will be closed, and the excavation will be complete.

We are now finished with the modeling. The model should appear as shown below.

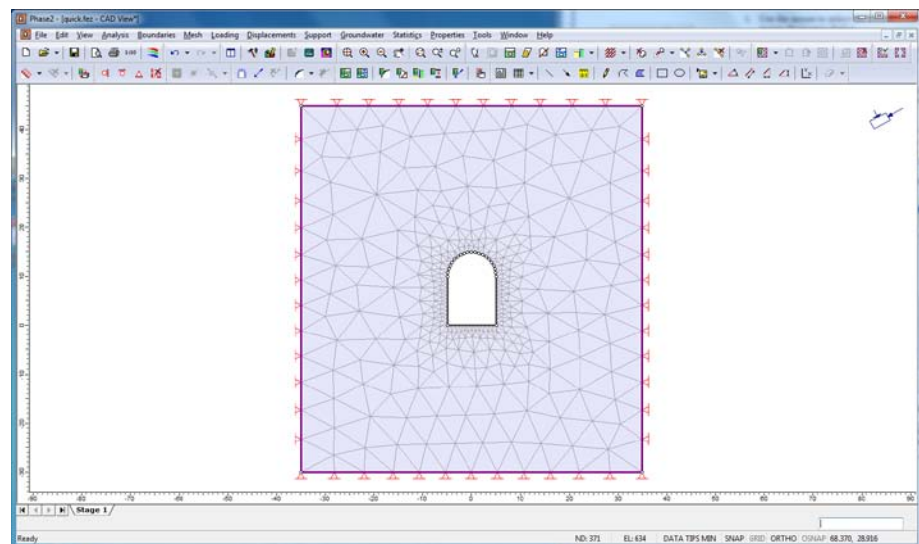


Figure 1-1: Finished model – Phase2 Quick Start Tutorial

TIP: assigning can also be done using a right-click shortcut (right-click in the desired area and the popup menu will have an Assign Material sub-menu).

Compute

Before you analyze your model, save it as a file called **quick.fez**. (*Phase2* files have an **.fez** filename extension.)



Select: File → Save

Use the Save As dialog to save the file. You are now ready to run the analysis.



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

To view the results of the analysis:



Select: Analysis → Interpret

This will start the *Phase2* Interpret program.

Principal Stress

By default, after a *Phase2* stress analysis, you will always see a contour plot of the major principal stress Sigma 1, when a file is opened in Interpret. This is shown in the figure below.

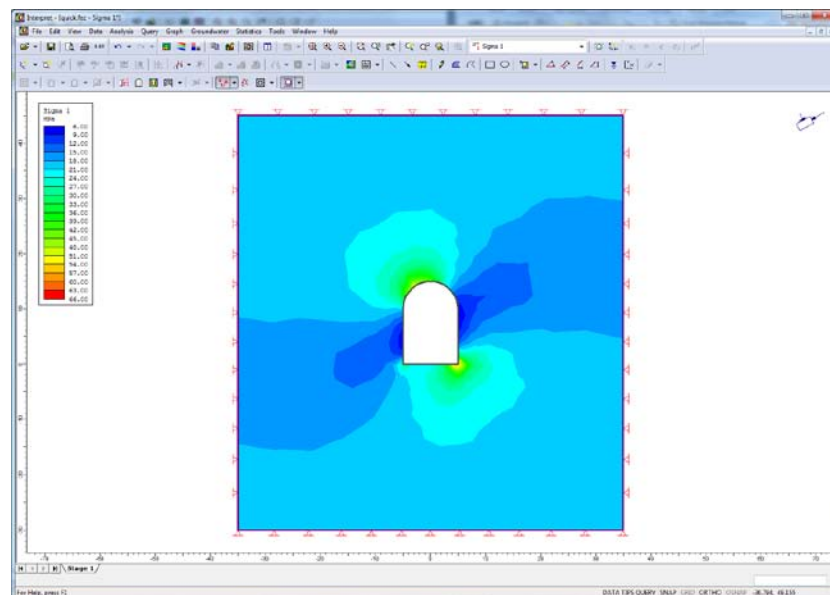


Figure 1-2: Contours of Major Principal Stress

Notice the effect of the field stress orientation (30 degrees from horizontal) on the Sigma 1 contours.

Now let's zoom in to get a closer look at the stress contours around the excavation. An easy shortcut to zoom in to your excavation(s), is to use the Zoom Excavation option.



Select: View → Zoom → Zoom Excavation

Notice the high stresses at the upper left and lower right of the excavation. The maximum Sigma 1 is at the sharp corner at the lower right.

Stress Trajectories



Now toggle the display of principal stress trajectories on, by selecting the **Stress Trajectories** toolbar button.

The principal stress trajectories are shown as small cross icons where the long axis of the cross is oriented in the direction of the major in-plane principal stress (Sigma 1) and the short axis is the direction of the minor in-plane principal stress (Sigma 3).

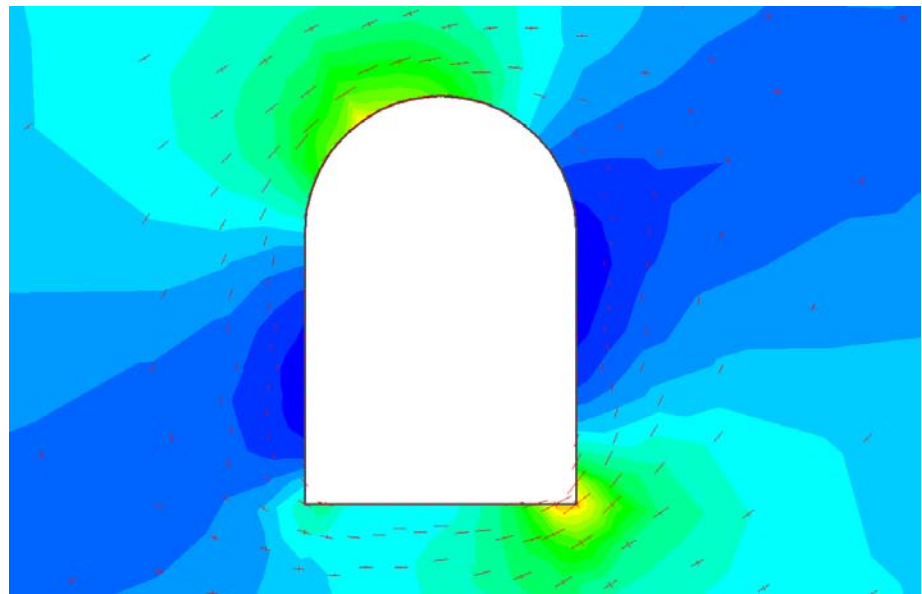


Figure 1-3: Display of principal stress trajectories

Toggle off the display of stress trajectories by re-selecting the Stress Trajectories toolbar button. (Stress Trajectories can also be turned on or off in the Display Options dialog.)

Creating a Query

We can graph the stress on the boundary or at any other location in the rock mass using the Query option. A Query allows you to view and plot analysis results based on the current contour data. (Queries are discussed in more detail in later tutorials).

1. Right-click on the excavation boundary and select **Query Boundary** from the popup menu.
2. Select OK in the dialog and a Query will be created for the boundary (you will see data values displayed along the boundary).
3. Right-click again on the excavation boundary, and select **Graph Data** from the popup menu.
4. Select **Plot** in the dialog, and you should see the following graph.

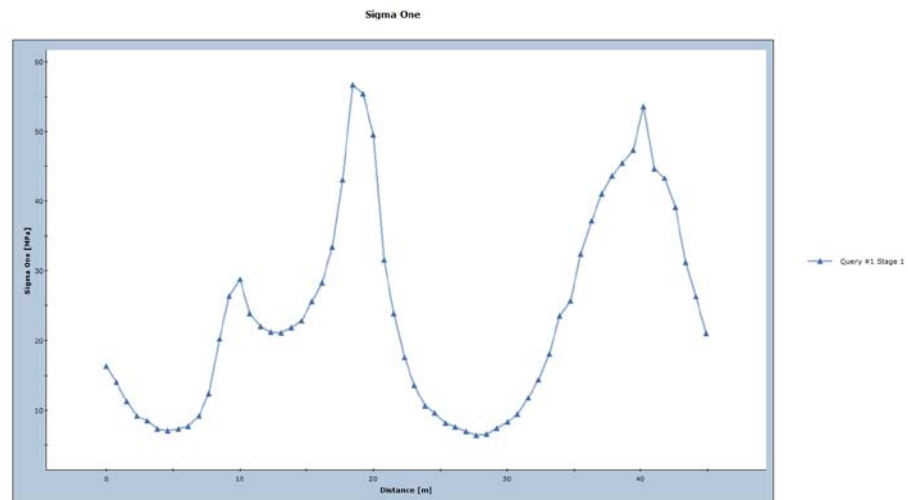


Figure 1-4: Principal stress on excavation boundary

The maximum stresses on the graph correspond to the maximum stresses on the Sigma 1 contour plot.

Close the query graph.

Right click on the excavation boundary and select the **Show Values** option, to turn off the display of the query values along the boundary. If you see the number 1 displayed on the boundary, this is the query ID number, which can also be turned on or off in the right-click menu.

Strength Factor

Let's now look at the Strength Factor contours. Select **Strength Factor** from the data list in the toolbar.

Select:  **Strength Factor**

Select Zoom All (or the F2 function key) to view the entire model.



Select: View → Zoom → Zoom All

Let's change the number of contour intervals, so that we get even numbered intervals.



Select: View → Contour Options

In the Contour Options dialog, select the Custom Range option and enter the Number (of contour intervals) = 7, and select Done. (Note: Contour Options is also available in the default right-click menu).

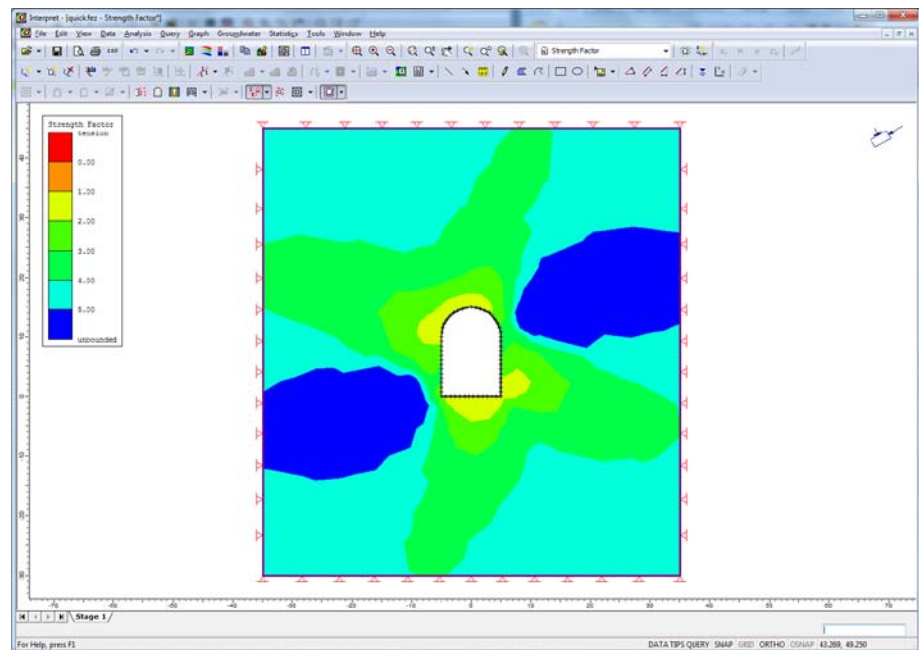


Figure 1-5: Strength Factor Contours.

Notice that the minimum strength factor contour interval is between 1 and 2. Therefore, based on this elastic analysis, no failure is to be expected for this model.

Let's verify this with the Query option. Right-click on the excavation boundary (which still has the query applied), and select **Graph Data** from the popup menu. You should see the following graph.

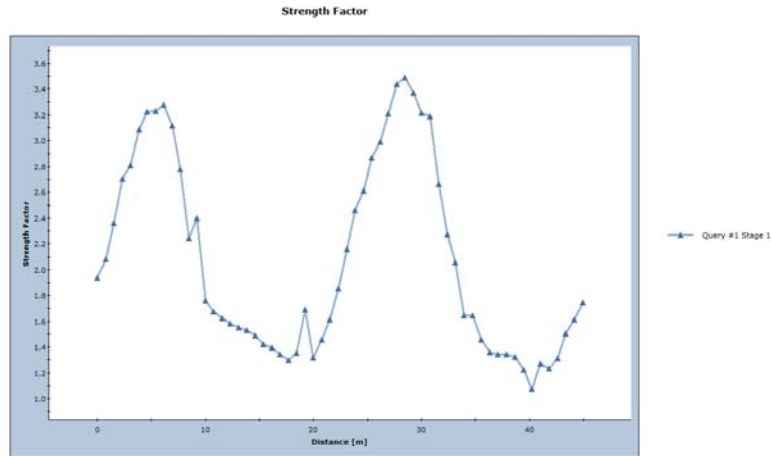


Figure 1-6: Strength Factor around excavation boundary.

As can be seen from the graph, the Strength Factor at the boundary is greater than 1 at all points (although it is close to 1 at two locations, which correspond to the high stress regions shown in Figure 1-4).

Because the strength factor is greater than 1 throughout the model, no additional information would be gained from a plastic analysis of this model (i.e. if you defined the rock mass material type as plastic, the analysis results would be the same).

Close the Strength Factor graph.

Right click on the excavation boundary and select **Delete Query** to remove the query from the model.

Displacements

Let's look at the displacements. Select **Total Displacement** from the data list in the toolbar.

Select:  **Total Displacement** ▼

The total displacement contours will be plotted, and the status bar will indicate the maximum displacement for the entire model (about 11 mm).

Maximum Total Displacement = 0.01143 m

Now select Zoom Excavation again.



Select: View → Zoom → Zoom Excavation

As can be seen from the contours, the maximum displacement is occurring at the excavation walls. Now let's display the deformation vectors and the deformed boundaries.



Select the Deformed Boundaries and Deformation Vectors buttons in the toolbar.



The deformed shape of the excavation boundaries is graphically illustrated by the use of these options. The deformation is magnified by a scale factor, which can be user-defined in the Display Options dialog. This is left as an optional exercise. In the figure below, a scale factor of 80 was used (i.e. displayed deformation = 80 times actual deformation).

Many of the Display Options are also available in the toolbar.

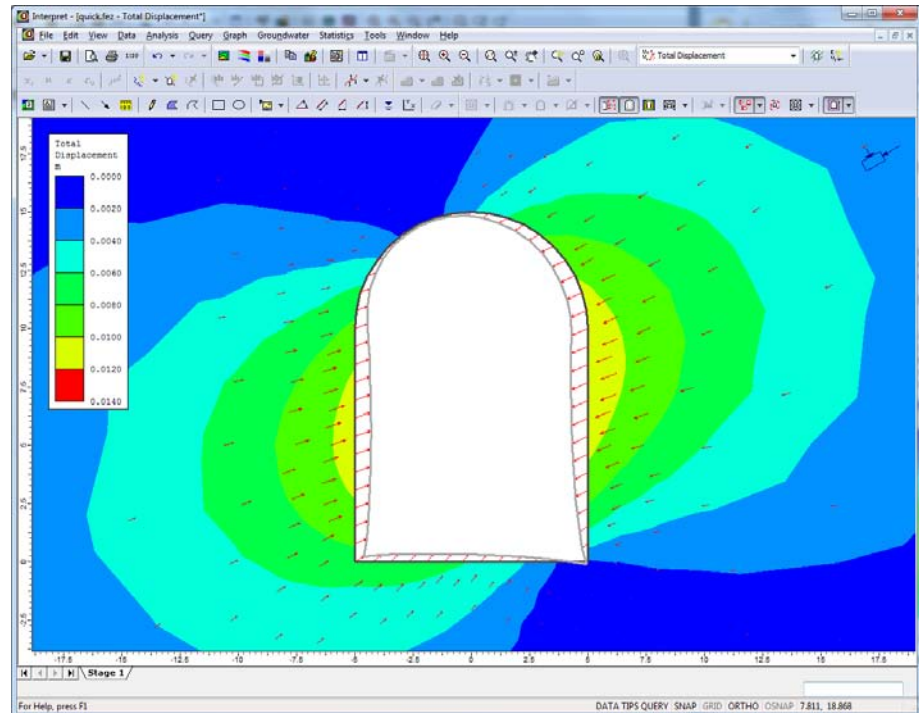


Figure 1-7: Total displacement contours, with deformation vectors and deformed boundaries displayed.

Toggle off the deformation vectors and deformed boundaries by re-selecting the corresponding buttons in the toolbar.

Contour Options is available in the right-click menu.

Now we will change the number of contour intervals, and add some contour labels. Right-click the mouse and select Contour Options.

In the Contour Options dialog, change the Number (of contour intervals) to 6. Select Done.

Contour Labels

Now let's add some labels to the contours, to identify the values represented by each contour boundary.

Select: Tools → Add Tool → Label Contour



A cross-hair cursor will appear on the screen. Click the left mouse button anywhere on or near a contour boundary, and a contour label will be placed at that point. The following figure illustrates what the display might look like after you have added some contour labels to the model. When you have added all of the labels that you wish, press the *Esc* key or right-click and select Cancel.

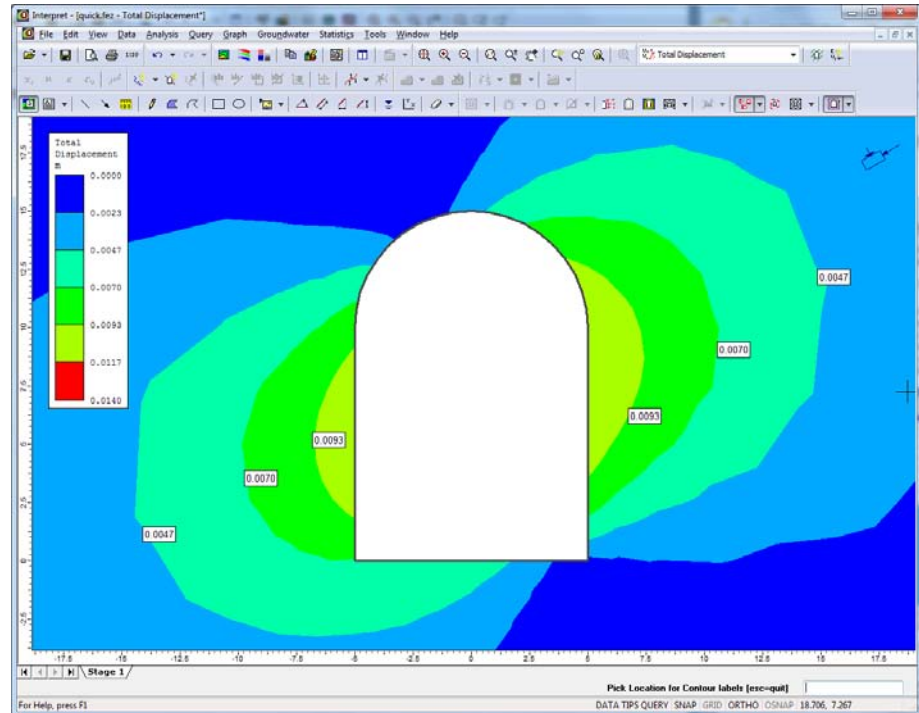


Figure 1-8: Contour labels added to displacement contour plot.

NOTE: the number of decimal places and number format used for the contour labels, can be customized in the Legend Options dialog. This is available in the View menu or by right-clicking on the Legend. The style (font size etc) used for the Contour Labels can be customized by double-clicking on a Contour Label.

Data Tips

A useful feature of *Phase2* are the popup Data Tips which allows the user to obtain model and analysis information, by simply placing the mouse cursor over any model entity or location on the screen.

To enable Data Tips, click on the box on the Status Bar (at the bottom of the *Phase2* application window), which says Data Tips. When you click on this box, it will toggle through 4 different Data Tip modes (including Off mode). Click on this box until it displays **Data Tips Max**.

DATA TIPS MAX SNAP GRID

Now move the mouse cursor over the model, and you will see that the material properties of the rock mass are displayed when the cursor is placed anywhere within the material. Place the cursor over the Stress Block in the upper right corner of the screen, and the Field Stress parameters will be displayed.

Click on the Data Tips box in the Status Bar, until it displays **Data Tips Query**. This mode allows you to obtain exact interpolated values of data at any point on the contour plots. Move the mouse around the contour plot, and notice that the exact value of the currently contoured variable is displayed, as well as the exact location coordinates.

DATA TIPS QUERY SNAP GRID

Click on the Status Bar and toggle **Data Tips Off**. Data Tips can also display a variety of other information, including support properties etc. The user is encouraged to experiment with this option in later tutorials.

Data Tips can also be toggled using the Data Tips sub-menu in the View menu.

Info Viewer

The **Info Viewer** option in the Analysis menu or the toolbar, displays a summary of *Phase2* input parameters and analysis results, in its own view.



Select: Analysis → Info Viewer

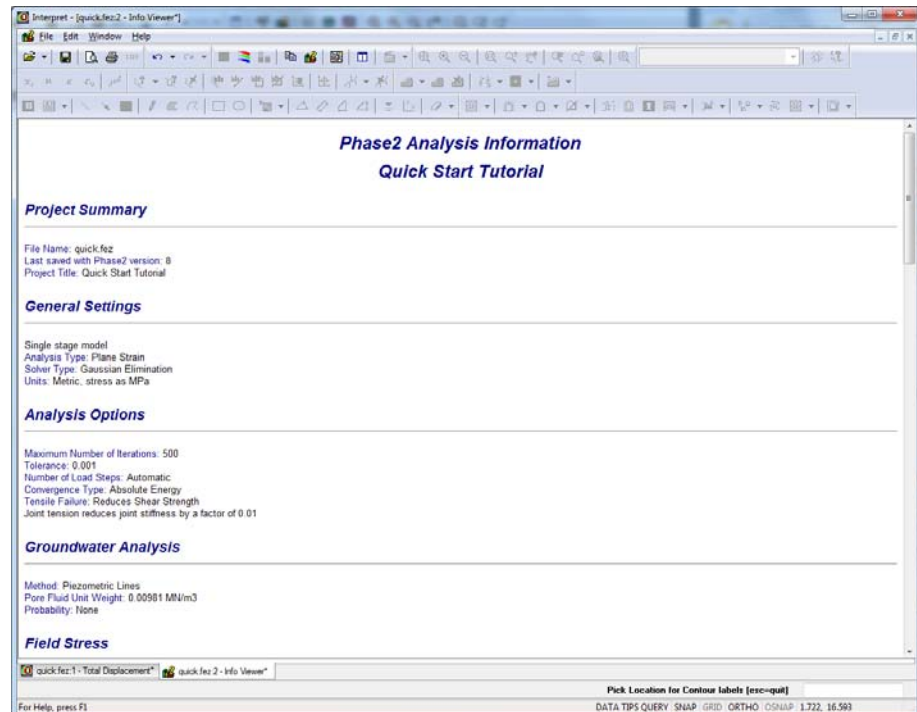


Figure 1-9: Phase2 Info Viewer listing.

The Info Viewer information can be copied to the clipboard using the Copy option in the toolbar or the Edit menu, or by right-clicking in the view and selecting Copy. From the clipboard, the information can be pasted into word processing programs for report writing.

The Info Viewer can also be saved to an html file using the Save option in the right-click menu or the File menu.

Close the Info Viewer view, by selecting the X in the upper right corner of the view.

Drawing Tools

In the **Tools** menu or the toolbar, a wide variety of drawing tools are available for customizing views. We will briefly demonstrate some of these options.

First, let's delete the contour labels we added previously.



Select: Tools → Delete All Tools

All contour labels will be deleted.

Now press F2 to Zoom All.

Let's add an arrow to the view. Select the **Arrow** option from the toolbar or the Tools menu.



Select: Tools → Add Tool → Arrow

Click the mouse at two points on the screen, to add an arrow pointing anywhere within the rock mass. Now let's add some text.



Select: Tools → Add Tool → Text Box

Click the mouse at a point near the tail of the arrow. You will see the **Add Text** dialog. The Add Text dialog allows you to type any text and add it to the screen. The Auto-Text option can be used to annotate the model with pre-formatted input and output data.

For example:

1. In the Add Text dialog, select the Material Properties “+” box (NOT the checkbox). Then select the Material: rock mass “+” box. Then select the Material: rock mass checkbox.
2. Now select the **Insert Auto-text** button. The Material Properties for the rock mass will be added to the editing area at the left of the Add Text dialog.

- Now select OK. The text is added to the view, and your screen should look similar to Figure 1-10.

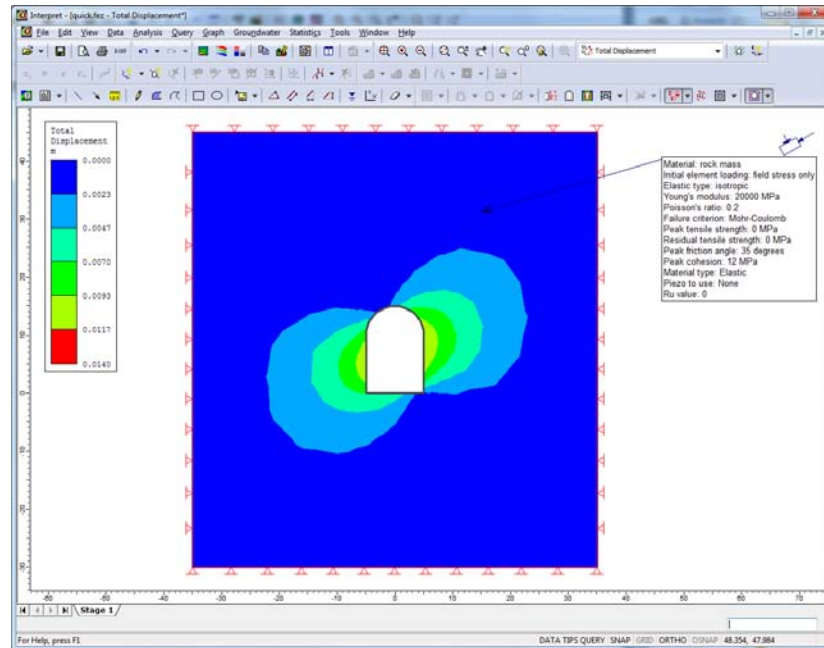


Figure 1-10: Auto-text and arrow added to view.

TIP: after adding a drawing tool, you can easily change the position, size or formatting style, by clicking on the tool with the mouse. This is described in the next section.

Many other drawing tools are available in *Phase2*, including options which allow you to add dimensioning, dynamic text, calculate areas of polygons, etc. The user is encouraged to experiment with the many different capabilities of the Drawing Tools in *Phase2*.

Editing Drawing Tools

We will now describe the following properties of all drawing tools added through the Tools menu options:

Right-click

If you *right-click the mouse* on a drawing tool, you will see a popup menu, which makes available various editing options.

For example:

- right-click on the arrow. Delete, Format and Copy options are available in the popup menu.
- right-click on the text box. Various options are available, including Delete, Format and Edit Text.

Single-click

If you *single-click the left mouse button* on a drawing tool, this will “select” the tool, and you will see the “control points” highlighted on the tool. While in this mode:

- You can click and drag the control points, to re-size the tool.
- If you hover the mouse over any part of the drawing tool, but NOT on a control point, you will see the four-way arrow cursor, allowing you to click and drag the entire drawing tool to a new location.
- You can delete the tool by pressing Delete on the keyboard.
- You can create a copy of the tool by pressing Ctrl-C on the keyboard, or by selecting Copy from the toolbar or the Edit menu.

Double-click

If you *double-click the mouse* on a drawing tool, you will see the Format Tool dialog. The Format Tool dialog allows you to customize styles, colours etc. Only the options applicable to the clicked-on tool, will be enabled in the Format Tool dialog. (Note: this is the same Format option available when you right-click on a tool).

It is left as an optional exercise, for the user to experiment with the various editing options that are available for each Tools option.

Saving Drawing Tools



To save drawing tools, select the **Save** option from the toolbar or the File menu. This will save all drawing tools, so that next time you open the file, the tools will re-appear on the view.

If you close a file in Interpret, you will automatically be prompted to save the drawing Tools, if you have not already saved them.

Exporting Images

In *Phase2*, various options are available for exporting image files.

Export Image File

The Export Image option in the File menu or the right-click menu, allows you to save the current view directly to one of four image file formats:

- JPEG (*.jpg)
- Windows Bitmap (*.bmp)
- Windows Enhanced Metafile (*.emf)
- Windows Metafile (*.wmf)

Copy to Clipboard



The current view can also be copied to the Windows clipboard using the Copy option in the toolbar or the Edit menu. This will place a bitmap image on the clipboard which can be pasted directly into word or image processing applications.

Black and White Images (Grayscale)



The Grayscale option, available in the toolbar or the View menu, will automatically convert the current view to Grayscale, suitable for black and white image requirements. This can be useful when sending images to a black and white printer, or for capturing black and white image files.

That concludes this 'quick start' tutorial. To exit the Interpret program:

Select: File → Exit